

1. Method 1: Model the trend + subtract it

This is the most intuitive approach.

Suppose your observed series is:

$$y_t = m_t + \epsilon_t$$

where:

- y_t = actual observed value at time t
- m_t = systematic trend
- ϵ_t = remaining/random component

If we can estimate m_t , then:

$$y_t - m_t = \epsilon_t$$

So the procedure is:

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Original time series

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Estimate the trend

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Subtract trend

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Detrended series

2. Linear trend

The simplest trend model is:

$$m_t = \alpha_0 + \alpha_1 t$$

This is just a straight line.

It is the same idea as ordinary linear regression:

$$y = \text{intercept} + \text{slope} \times x$$

Here:

- α_0 = intercept
- α_1 = slope
- t = time

Example

Suppose we have:

Time t	Actual y_t
1	12
2	14
3	15
4	18
5	19

Suppose we estimate:

$$m_t = 10 + 1.8t$$

Then:

t	Actual y_t	Estimated trend m_t
1	12	11.8
2	14	13.6
3	15	15.4
4	18	17.2
5	19	19.0

Now subtract:

$$y_t - m_t$$

For $t = 1$:

$$12 - 11.8 = 0.2$$

For $t = 2$:

$$14 - 13.6 = 0.4$$

For $t = 3$:

$$15 - 15.4 = -0.4$$

So the detrended series is approximately:

$$[0.2, 0.4, -0.4, 0.8, 0]$$

The large upward movement has been removed.

3. What does α_0 mean?

In:

$$m_t = \alpha_0 + \alpha_1 t$$

α_0 is the intercept.

It represents the estimated trend level when:

$$t = 0$$

For example:

$$m_t = 10 + 1.8t$$

means:

$$\alpha_0 = 10$$

4. What does α_1 mean?

α_1 is the slope.

It tells us how much the trend changes for one unit of time.

For:

$$m_t = 10 + 1.8t$$

we have:

$$\alpha_1 = 1.8$$

So the estimated trend increases by **1.8 units per time period**.

If:

$$\alpha_1 > 0$$

→ increasing trend.

If:

$$\alpha_1 < 0$$

→ decreasing trend.

If:

$$\alpha_1 = 0$$

→ no linear trend.

5. How do we find α_0 and α_1 ?

Usually, we estimate them using **linear regression / least squares**.

We fit:

$$y_t = \alpha_0 + \alpha_1 t + \epsilon_t$$

The regression finds the line that best represents the systematic movement in the data.

Then:

$$\hat{m}_t = \hat{\alpha}_0 + \hat{\alpha}_1 t$$

The hat $\hat{}$ means estimated.

Then:

$$y_t - \hat{m}_t$$

gives the detrended series.

6. Quadratic trend

What if the trend isn't a straight line?

This is exactly what you saw in your previous graph.

Your quadratic trend was:

$$m_t = \alpha_0 + \alpha_1 t + \alpha_2 t^2$$

Now we have three parameters:

Parameter	Meaning	 
α_0	Intercept	
α_1	Linear component	
α_2	Curvature	

The t^2 term allows the trend to **bend**.

For example:

$$m_t = 2 + 0.5t + 0.05t^2$$

Initially the trend increases slowly, but the t^2 component causes the increase to become stronger over time.

7. Connecting this directly to your previous code

Your previous DGP was:

$$y_t = 0.1 + 0.07t + 0.001t^2 + \epsilon_t$$

Therefore:

$$m_t = 0.1 + 0.07t + 0.001t^2$$

is the trend.

The noise is:

$$\epsilon_t$$

So if we know the trend, we can calculate:

$$y_t - m_t$$

which gives:

$$\begin{aligned} y_t - m_t &= (0.1 + 0.07t + 0.001t^2 + \epsilon_t) \\ &\quad - (0.1 + 0.07t + 0.001t^2) \\ &= \epsilon_t \end{aligned}$$

That's the perfect theoretical example of detrending.

8. Method 2: Differencing

Now we have a completely different approach.

Instead of explicitly estimating the trend, we calculate:

$$\Delta y_t = y_t - y_{t-1}$$

This is called the **first difference**.

In simple words:

Compare today's value with yesterday's value.

Example

Suppose:

Time	y_t
1	100
2	105
3	110
4	116
5	114

First difference:

At $t = 2$

$$\Delta y_2 = 105 - 100 = 5$$

At $t = 3$

$$\Delta y_3 = 110 - 105 = 5$$

At $t = 4$

$$\Delta y_4 = 116 - 110 = 6$$

At $t = 5$

$$\Delta y_5 = 114 - 116 = -2$$

So:

$$\Delta y = [5, 5, 6, -2]$$

Instead of modeling the **level**, we're modeling the **change**.

9. Why does differencing remove a linear trend?

This is a beautiful result.

Suppose:

$$y_t = \alpha_0 + \alpha_1 t + \epsilon_t$$

Then:

$$y_{t-1} = \alpha_0 + \alpha_1(t-1) + \epsilon_{t-1}$$

Now calculate:

$$\Delta y_t = y_t - y_{t-1}$$

Substitute:

$$\Delta y_t = (\alpha_0 + \alpha_1 t + \epsilon_t) - (\alpha_0 + \alpha_1(t-1) + \epsilon_{t-1})$$

Expand:

$$= \alpha_0 + \alpha_1 t + \epsilon_t - \alpha_0 - \alpha_1 t + \alpha_1 - \epsilon_{t-1}$$

Everything cancels except:

$$\boxed{\Delta y_t = \alpha_1 + \epsilon_t - \epsilon_{t-1}}$$

Notice what happened.

The original trend:

$$\alpha_0 + \alpha_1 t$$

has disappeared.

That's why differencing is useful for removing a linear/stochastic trend.

10. A very intuitive way to understand differencing

Imagine:

Original:

100

105

110

115

120

125

There is a strong upward trend.

Take differences:

$$105 - 100 = 5$$

$$110 - 105 = 5$$

$$115 - 110 = 5$$

$$120 - 115 = 5$$

$$125 - 120 = 5$$

You get:

5

5

5

5

5

The trend in the original data has become a **constant level** in the differenced data.

If there is noise:

Original:

100

106

109

116

119

127

Differences:

6

3

7

3

8

Now we're looking at the **changes**, rather than the steadily increasing level.

Detrending vs Differencing

Both are used to deal with non-stationary time series, but they remove the trend in different ways.

1. Detrending: model and subtract the trend

In detrending, we explicitly estimate what the trend looks like and subtract it from the original series.

Suppose:

$$y_t = m_t + \epsilon_t$$

where m_t is the trend.

For a linear trend:

$$m_t = \alpha_0 + \alpha_1 t$$

Then:

$$y_t^{\text{detrended}} = y_t - \hat{m}_t$$

Example:

Original: 10 12 15 17 20

Trend: 9 11 13 15 17

Detrended: 1 1 2 2 3

So detrending asks:

"What is the underlying trend, and what remains after removing it?"

It is especially useful when you can reasonably describe the trend using a **linear, quadratic, or other functional form**.

2. Differencing: calculate change between consecutive observations

Differencing does not explicitly model the trend.

Instead, it calculates:

$$\Delta y_t = y_t - y_{t-1}$$

Example:

Original:

100 → 105 → 110 → 116 → 120

Difference:

5 5 6 4

So differencing asks:

"How much did the series change from one period to the next?"

For a linear trend:

$$y_t = \alpha_0 + \alpha_1 t + \epsilon_t$$

first differencing removes the time-dependent linear component:

$$\Delta y_t = y_t - y_{t-1}$$

and gives approximately:

$$\Delta y_t = \alpha_1 + \epsilon_t - \epsilon_{t-1}$$

So the level/trend is removed, leaving the changes.

12. Method 3: High-pass filtering

This sounds complicated, but the intuition is quite simple.

A time series can contain movements at different **frequencies**.

Think of:

Very slow movement

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Trend

Medium movement

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Cycles

Fast movement

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Random fluctuations

A **filter** allows some types of movement to pass through and suppresses others.

13. Low frequency vs high frequency

Low-frequency movement

Changes slowly over time:

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This could represent a long-term trend or cycle.

High-frequency movement

Changes rapidly:

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This could represent short-term fluctuations/noise.

A **high-pass filter** emphasizes the faster movements and removes/suppresses the slower movements.

Conceptually:

Original series

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↓

High-pass filter

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+---- Low-frequency trend → suppressed

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+---- High-frequency variation → retained

14. Hodrick-Prescott filter

The Hodrick-Prescott (HP) filter is commonly used to separate a time series into:

$$y_t = \tau_t + c_t$$

where:

- y_t = observed series
- τ_t = smooth trend
- c_t = cyclical component

So:

$$c_t = y_t - \tau_t$$

The HP filter tries to find a trend that is:

1. close to the observed data
2. sufficiently smooth

Its optimization problem is:

$$\min_{\tau_t} \sum_t (y_t - \tau_t)^2 + \lambda \sum_t [(\tau_{t+1} - \tau_t) - (\tau_t - \tau_{t-1})]^2$$

You don't need to memorize this immediately.

The intuition is more important.

15. What does λ do?

The HP filter has a parameter called:

$$\lambda$$

which controls how smooth the estimated trend should be.

Small λ

The trend is allowed to follow the data more closely.

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Data: /_/_/_/_/_/\

Trend: /~~~~~\~~~~~\~~~~~\~~~~~\~~~~~\~~~~~\~~~~~\

Large λ

The trend is forced to be smoother.

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Data: /_/_/_/_/_/\

Trend: /~~~~~\~~~~~\~~~~~\~~~~~\~~~~~\~~~~~\~~~~~\

So:

$\lambda \uparrow \Rightarrow$ smoother trend

and:

$\lambda \downarrow \Rightarrow$ more flexible trend

16. Why is HP called a filtering approach?

Because instead of saying:

"My trend is definitely linear."

or:

"My trend is definitely quadratic."

we allow the data to determine a **smooth trend**.

Compare:

Linear model

We assume:

$$m_t = \alpha_0 + \alpha_1 t$$

Quadratic model

We assume:

$$m_t = \alpha_0 + \alpha_1 t + \alpha_2 t^2$$

HP filter

We say:

"Find a smooth trend that follows the data reasonably well, while avoiding excessive wiggles."

That's much more flexible.

17. All three methods together

Method	What you do	Main idea
Model + subtract	Estimate m_t , then $y_t - m_t$	Explicitly model trend
Differencing	$y_t - y_{t-1}$	Remove changes that persist across time
HP filter	Estimate smooth τ_t , then $y_t - \tau_t$	Separate smooth trend from shorter-term movement

To remove a **long-term pattern (trend)** from a time series, you generally **estimate the trend and subtract it from the original data**.

1. The basic idea

Suppose your original time series is:

$$Y_t = Trend_t + ShortTerm_t$$

Estimate the trend:

$$\hat{T}_t$$

Then calculate:

$$Detrended_t = Y_t - \hat{T}_t$$

The result should fluctuate around **zero**, like the bottom graph in your image.

2. In your example

Your graph has a clear upward trend. You could fit a **linear** or **quadratic** trend:

$$\hat{T}_t = a + bt$$

or

$$\hat{T}_t = a + bt + ct^2$$

Then subtract it:

Now detrended contains the fluctuations after removing the long-term trend.

Deseasonalizing means removing the seasonal pattern from a time series.

A **seasonal pattern** is a pattern that repeats at a fixed interval.

For example, suppose monthly ice-cream sales are:

Month Sales

Jan 100

Feb 110

Mar 120

... ..

Jun 180

Month Sales

Jul 190

... ..

Dec 120

If sales are **consistently higher every summer**, that repeating pattern is **seasonality**.

Deseasonalizing

We separate:

$$Y_t = \text{Trend} + \text{Seasonality} + \text{Random error}$$

Then remove the seasonal component:

$$Y_t^{\text{deseasonalized}} = Y_t - \text{Seasonal Component}$$

For a multiplicative model:

$$Y_t^{\text{deseasonalized}} = \frac{Y_t}{\text{Seasonal Index}}$$

Method	What is removed?	Purpose
Detrending	Long-term upward/downward movement	Remove trend
Deseasonalizing	Repeating seasonal pattern	Remove seasonality
Differencing	Previous-period level	Focus on changes

Easy memory:

- **Detrending** → remove the long-term direction ↗
- **Deseasonalizing** → remove the repeating pattern 🔄
- **Differencing** → calculate the change from one period to the next

Example: If every December sales increase because of Christmas, **deseasonalizing removes that predictable December effect** so you can study the underlying movement more clearly.

3. What is AR(1)?

AR means **AutoRegressive**.

AR(1) means:

Today's value depends on yesterday's value + a new random shock.

The mathematical equation is:

$$Y_t = \phi Y_{t-1} + \epsilon_t$$

For your code:

$$Y_t = 0.7Y_{t-1} + \epsilon_t$$

where:

- Y_t = current value
- Y_{t-1} = previous value
- 0.7 = AR coefficient
- ϵ_t = random noise/shock

Example

Suppose:

$$Y_{t-1} = 10$$

and today's random shock is:

$$\epsilon_t = 2$$

Then:

$$Y_t = 0.7(10) + 2$$

$$Y_t = 9$$

So today's value is influenced by yesterday's value.

This plot shows a **time series generated by an AR(1) model** — an **autoregressive model of order 1**.

What AR(1) means

An AR(1) model assumes that the current value depends partly on the **previous value**, plus some random noise:

$$X_t = c + \phi X_{t-1} + \epsilon_t$$

where:

- X_t = value at time t
- X_{t-1} = value at the previous time step
- ϕ = **AR(1) coefficient**
- ϵ_t = random error/noise

How to read your graph

- **X-axis (0–500):** time/observation number.
- **Y-axis:** value of the time series.
- The orange line is the value X_t at each time.
- The values fluctuate roughly between **–3.5 and +4.5**.
- There doesn't appear to be a sustained upward or downward trend.
- The series repeatedly moves around a central level (roughly 0).

The important feature of an AR(1) process is **autocorrelation**: consecutive observations are related.

For example, if $\phi > 0$:

$$X_t \text{ tends to be similar to } X_{t-1}$$

So a positive value tends to be followed by another positive-ish value, and a negative value by another negative-ish value.

If ϕ is close to **0**, the observations behave more like independent random noise. If ϕ is close to **1**, the series tends to be much more persistent.

One thing we can't tell from this graph alone

The legend says **AR(1)**, but the actual value of the AR coefficient ϕ isn't shown. So we can't determine exactly how strong the dependence is just by looking at the plot.

If you give me the **AR(1) code you used to generate this graph**, I can explain exactly what each part means and why the graph looks like this.